Template Attacks on Masking—Resistance Is Futile*

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Abstract. In this article we discuss different types of template attacks on masked implementations. All template attacks that we describe are applied in practice to a masked AES software implementation on an 8-bit microcontroller. They all break this implementation. However, they all require quite a different number of traces. It turns out that a template-based DPA attack leads to the best results. In fact, a template-based DPA attack is the most natural way to apply a template attack to a masked implementation. It can recover the key from about 15 traces. All other attacks that we present perform worse. They require between about 30 and 1800 traces. There is no difference between the application of a template-based DPA attack to an unmasked and to a masked implementation. Hence, we conclude that in the scenario of template attacks, masking does not improve the security of an implementation.

1 Introduction

Power analysis attacks and countermeasures are among the most active areas of research in applied cryptography. In this field scientists investigate the practical security of implementations of cryptographic algorithms. In particular, they study how the information that leaks through the power consumption of a cryptographic device can be exploited in practice. It has turned out that unprotected devices, no matter which cryptographic algorithm they implement, are an easy target for power analysis attacks.

In the pioneering work of Kocher *et al.* [KJJ99], simple power analysis (SPA) and differential power analysis (DPA) attacks have been introduced. SPA attacks have been described as attacks in which the attacker has access to only a very small set of power traces of a devices. In contrast, DPA attacks, are based on a large number of power traces. Consequently, the description of DPA attacks includes a statistical method that can be used to determine the unknown key.

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A particularly interesting class of power analysis attacks, so-called template attacks, has been introduced by Chari *et al.* [CRR03]. In a template-based power analysis attack, the attacker is assumed to know the power consumption characteristics of some instructions of a device. This characterization is stored and called template. Templates are then used as follows. In a template-based DPA attack, the attacker matches the templates based on different key hypotheses with the recorded power traces. The templates that match best indicate the key. This type of attack is the best attack in an information theoretic sense, see [CRR03].

A very popular countermeasure to defeat power analysis attacks is called masking. In a masked implementation, all intermediate values that are computed during the execution of a cryptographic algorithm are concealed by a random value, which is called mask. This ensures, if correctly implemented, a pairwise independence between masked values, unmasked values, and masks. As a consequence, (first-order) DPA attacks do not work.

It has been observed previously, that masked implementations can be attacked by second-order DPA attacks and template-based DPA attacks. However, the research in this area is scattered. Several groups have applied different concepts of attacks to different types of implementations making different assumptions. Consequently, by reading the so-far published literature it is hard to evaluate which attack strategy works best. A related question is how many power traces does a template-based DPA attack need to break a masked implementation? And, continuing this line of thought: is there any difference in the security between an unmasked and a masked implementation in the template scenario?

In this article, we aim to answer these questions. We explain several ways to apply template attacks to masked implementations of block ciphers. These theoretical descriptions are followed by practical results of the implementation of the described attacks. This means, we have implemented all attacks and applied them to an implementation of a masked AES on an 8-bit microcontroller. This allows a fair comparison of the attacks on software implementations on a typical smartcard processor. Our research leads to devastating conclusions for the security of masked implementations of block ciphers on smartcards. In the scenario of template attacks, there is actually no difference in the security of masked and unmasked implementations. Template attacks work virtually in the same way and have the same effectiveness. They can recover the key from about 15 power traces.

This article is organized as follows. Sect. 2 reviews template attacks and discusses related work. In Sect. 3 we explain several template attacks to break masked implementations. In Sect. 4 we apply these attacks to a masked AES software implementation on a microcontroller. We conclude our research in Sect. 5.

2 Template Attacks

The beauty of a standard DPA attack, such as described in [KJJ99], is that it can be conducted by an attacker with relatively low effort. This means, in